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Enhancing Quantum NLP Robustness – Analysis on Noisy Models for Quantum Sentiment Classification

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***Abstract*—This research aims to improve the robustness of Quantum Natural Language Processing (QNLP) models in the presence of noise. Using Lambeq, an open-source library for quantum NLP, we focus on sentiment classification tasks to evaluate the impact of noisy quantum models. Our study involves the application of three types of optimizers: Simultaneous Perturbation Stochastic Approximation (SPSA), Nelder-Mead, and Adam. The performance and resilience of these optimizers are assessed under various noise conditions to determine their effectiveness in maintaining model accuracy and stability. The research shows that the Adam optimizer outperforms SPSA and NelderMead in QNLP simple sentiment analysis tasks, both in noiseless and noisy environments. Adam achieved 96.67% accuracy in noiseless conditions, significantly higher than SPSA's 83.33% and Nelder-Mead's 53.33%, highlighting its superior performance in optimizing quantum circuits. Moreover, in noisy simulation, Adam outperforms others with 70.00% accuracy performance. The work is useful for NLP implementations in quantum computers. This research is essential because the noisy nature of current quantum computers makes it difficult for QNLP models to perform accurately and reliably.**

***Index Terms*—Quantum Natural Language Processing, sentiment classification, noisy quantum, robustness**

I. Introduction

The rapid advancement of quantum computing has paved the way for innovative applications in various domains, one of which is Quantum Natural Language Processing (QNLP). QNLP leverages quantum computing principles to enhance the processing and understanding of human language, promising significant improvements over classical NLP methods [1]. Despite its potential, QNLP faces several challenges, one of the most critical being the impact of noise on quantum operations [2]. Quantum computers, unlike their classical counterparts, are highly susceptible to noise due to their delicate quantum states and interactions with the environment. Noise can introduce errors, leading to reduced accuracy and reliability of quantum computations. This issue is particularly pertinent in QNLP tasks where precision and robustness are paramount. Therefore, understanding and mitigating the effects of noise is crucial for the practical implementation of QNLP models.

The primary objective of this research is to enhance the robustness of QNLP models in the presence of noise. By focusing on sentiment classification tasks using Lambeq, an open-source library for quantum NLP, we aim to evaluate how different optimization techniques can mitigate the adverse effects of noise [3]. Specifically, we investigate three types of optimizers: Simultaneous Perturbation Stochastic Approximation (SPSA), Nelder-Mead, and Adam. Through comprehensive experiments and analysis, this study seeks to identify optimization strategies that enhance the resilience and performance of QNLP models under noisy conditions.

The organization of the paper is as follows: Section I introduces the paper. Section II describes the related works of QNLP, sentiment analysis, and quantum error. Section III explains the basics needed for the proposed method. Section IV shows the proposed method and Section V shows the experimental results. Finally, Section VI concludes the paper.

II. Related Works

A. Methods and Tools for Quantum Natural Language Processing (QNLP)

In this study, we apply Lambeq [4], a high-level Python library specifically created for Quantum Natural Language Processing (QNLP). This open-source toolkit offers classes and modules that enable the creation of quantum circuits, tensor networks, and string diagrams that can be executed on a quantum computer.

Subsequent studies [5], such as those by Lorenz et al. (2023), have successfully trained NLP models on noisy quantum hardware for simple sentence classification tasks, showing that NISQ computers can handle datasets with over 100 sentences.

B. Quantum Approach for Sentiment Classification using Optimizer

Previous research has demonstrated the effectiveness of training QNLP models in noisy quantum environments, such as implementations of NLP tasks on NISQ hardware [6], which

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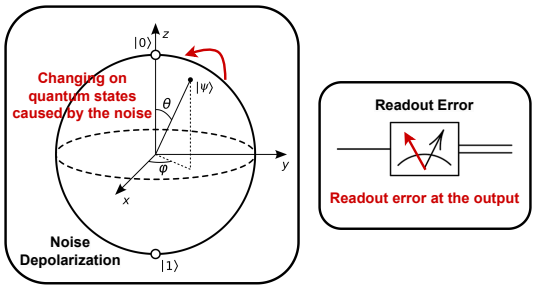


Fig. 1: Noise Depolarization and Readout Error in Quantum

utilized the DisCoCat model and SPSA for optimization. This approach showed promising results in handling noise during the training process, particularly in tasks such as question answering using small datasets.

C. Quantum Error

Previous work on quantum error correction has primarily focused on mitigating the impact of various types of errors that occur in quantum systems. Efforts have been made to address issues such as depolarizing errors, which cause a loss of coherence in quantum states, in quantum simulations [7]. There also have been efforts to understand readout errors, which cause inaccuracies in the measurement of qubits, by classifying correlations of readout errors on near-term quantum computers [8]. Despite significant efforts to mitigate errors in quantum systems, such errors continue to occur, posing challenges to the reliability and accuracy of quantum computations.

III. Theoretical Background

In this section, we present the key theoretical foundations necessary for readers to comprehend our proposed methods. Specifically, we explore five main topics: Sentiment Classification, Simultaneous Perturbation Stochastic Approximation, Nelder-Mead, Adam, and hybrid quantum-classical machine learning.

A. Sentiment Classification

Sentiment analysis (SA) involves analyzing opinions or sentiments expressed in natural language, typically categorized as positive, negative, neutral, or varying degrees in between [9]. It plays a crucial role in understanding public perception towards products, services, or content, evident in customer reviews, social media comments, and news sentiments.

B. Hybrid Quantum-Classical Machine Learning

A hybrid quantum-classical training algorithm for variational circuits leverages both quantum and classical computing resources to optimize circuit parameters [10]. The quantum device is employed to calculate elements of an objective function or its derivatives, providing information that the classical device uses to adjust the circuit parameters.

Algorithm 1: SPSA Optimization Algorithm

Input: N: Maximum number of iterations, Mag: Initial gradient estimate, α, γ: Decay rates for a and c, respectively, R: Random number, $L(\theta)$: Loss function, W: Parameter vector

Output: Updated W

begin

- $A \leftarrow 0.001 \times N$
- $a \leftarrow \frac{0.1(A+1)^{\alpha}}{Mag}$
- **for** $k = 1, 2, \ldots, N$ **do**
  - $a_k \leftarrow \frac{a}{(k+A)^{\alpha}}$
  - $c_k \leftarrow \frac{c}{k^{\gamma}}$
  - $\delta_k \leftarrow \epsilon$
  - $g \leftarrow L(w_k + c_k\delta_k) - L(w_k - c_k\delta_k)$
  - $w_k \leftarrow w_k - a_k g$
- return W

C. Simultaneous Perturbation Stochastic Approximation Optimizer

The Simultaneous Perturbation Stochastic Approximation (SPSA) is a method used to optimize a function, especially when the objective function is noisy and the exact gradient is unknown, difficult, or expensive to compute [11]. The SPSA algorithm aims to estimate the gradient using only two measurements of the objective function, even in high-dimensional spaces.

SPSA uses a Bernoulli-like distribution for each of the components of the parameter vector w. Every variable has an equal likelihood of being either 1 or -1 due to the distribution [12]. The addition results in the w_{plus} vector, whereas the subtraction results in the w_{minus} vector. The cost function is then evaluated at the point defined by these two vectors, and the slope or gradient is approximated. The following is the equation that represent the context above:

$$\frac{\partial L}{\partial w} = \frac{(L(w_{plus}) - L(w_{minus}))}{(2 \times A_p)},$$

where L is the cost function, w_{plus} is a vector containing the addition results, w_{minus} is a vector containing the subtraction results, and A_p is the perturbation amplitude. Alg. 1 shows the Simultaneous Perturbation Stochastic Approximation (SPSA) algorithm.

D. Nelder-Mead Optimizer

The Nelder-Mead optimizer, also referred to as the simplex method, is a very well-known derivative-free optimization technique applied to nonlinear optimization problems [13]. It works particularly well while dealing with smooth objective functions that might contain multiple local minima. This method maintains a simplex formed from $n+1$ points in an N-dimensional

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space, whereas N is the number of variables. Alg. 2 shows the pseudocode of Nelder-Mead Optimizer.

Algorithm 2: Nelder-Mead Optimization Algorithm

Input: $f(x)$: Objective Functions, X: Initial parameter points
Output: X: Optimal parameter point
begin
  Initialize the simplex with points $X \leftarrow [x_1, x_2, \ldots, x_{N+1}]$
  for $k = 1, 2, \ldots, N+1$ **do**
    Evaluate the objective function at each point of the simplex
    Order the points such that $f(x_1) \leq f(x_2) \leq \ldots \leq f(x_N)$
    Compute the centroid x_c of the best N points
    Reflect $x_r \leftarrow x_c + \alpha(x_c - x_{N+1})$
    if $f(x_r) < f(x_1)$ **then**
      Expand $x_e \leftarrow x_c + \gamma(x_r - x_c)$
      if $f(x_e) < f(x_r)$ **then**
        Replace x_{N+1} with x_e
      else
        Replace x_{N+1} with x_r
    else if $f(x_r) < f(x_N)$ **then**
      Replace x_{N+1} with x_r
    else
      if $f(x_r) < f(x_{N+1})$ **then**
        Contract outside $x_o \leftarrow x_c + \beta(x_r - x_c)$
        if $f(x_o) < f(x_r)$ **then**
          Replace x_{N+1} with x_o
        else
          Shrink the simplex towards x_1
      else
        Contract inside $x_i \leftarrow x_c - \beta(x_{N+1} - x_c)$
        if $f(x_i) < f(x_{N+1}$ **then**
          Replace x_{N+1} with x_i
        else
          Shrink the simplex towards x_1
  return X

The algorithm keeps on improving this simplex by computing the objective function at its vertices and updating the vertex that turns out to be the worst as viewed from the objective function by a better point as its alternate choice. Its principal operations can be reflection, expansion, contraction, or shrinking of the simplex [14]. These operations help in searching the space and ultimately converge into a minimum.

E. Adam Optimizer

Adam is a stochastic optimization with adaptive learning rates. Some of Adam's advantages are that the magnitudes of parameter updates are invariant to rescaling of the gradient, its stepsizes are approximately bounded by the stepsize hyperparameter, it does not require a stationary objective, it works with sparse gradients, and it naturally performs a form of step size annealing [15]. Alg. 3 shows the pseudocode of the Adam Optimizer. The hyperparameters for this optimizer are as follows.

1) Learning Rate (α), the step size for updating the weights.
2) Beta 1 (β_1), the average of gradients.
3) Beta 2 (β_2), the variance of gradients.
4) Epsilon (ϵ), a small constant to prevent division by zero. The variables processed for this optimizer are as follows.
5) Gradient (g_t), The partial derivatives of the loss function concerning the weights.
6) First Moment Vector (m_t), The exponentially decaying average of past gradients.
7) Second Moment Vector (v_t), The exponentially decaying average of past squared gradients.
8) Bias-Corrected Estimates (θ_t), to correct initial bias.

Algorithm 3: Adam Optimization Algorithm

Input: $f_t(\theta)$: Stochastic objective function, α: Stepsize, $\beta_1, \beta_2 \in [0, 1)$: Exponential decay rates, $f(\theta)$: Stochastic objective function, θ_0: Initial parameter vector
Output: Final parameter vector θ_t
begin
  $m_0 \leftarrow 0$
  $v_0 \leftarrow 0$
  $t \leftarrow 0$
  while θ_t *is not converged* **do**
    $t \leftarrow t + 1$
    $g_t \leftarrow \nabla_0 f_t(\theta_{t-1})$
    $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$
    $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$
    $\hat{m}_t \leftarrow \frac{m_t}{1 - \beta_1^t}$
    $\hat{v}_t \leftarrow \frac{v_t}{1 - \beta_2^t}$
    $\theta_t \leftarrow \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$
  return θ_t

IV. Methodology and Experiment Setups

A. Experimental Setup

This experiment uses a general Quantum Natural Language Processing (QNLP) pipeline tailored for sentiment classification tasks, as depicted in Fig. 2. This pipeline encompasses multiple stages, each transforming the input through distinct processing steps until the final evaluation metrics are obtained. The basics of the QNLP for sentiment analysis are in [16], [17].

Initially, the pipeline begins with sentences, which are the raw textual data that require analysis. The first transformative

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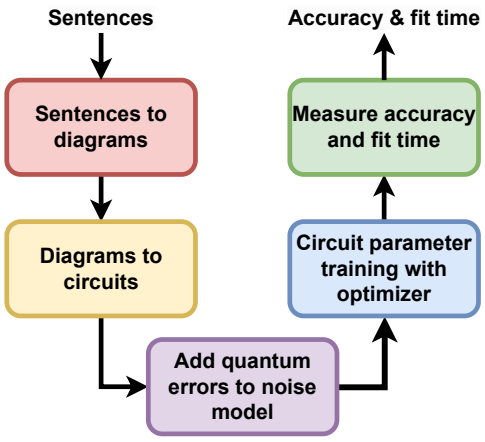


Fig. 2: General QNLP Pipeline for sentiment analysis with noise analysis

step converts these sentences to diagrams. This step involves mapping the syntactic and semantic structure of sentences into a graphical representation, using the DisCoCat (Distributional Compositional Categorical) model that fuses grammatical structure with vector-based meaning representations. Next, these diagrams are converted to circuits. In this stage, the graphical representations are translated into quantum circuits. In this work, we use Lambeq's IQP anzatz to represent the quantum circuits [16]. Once the circuits are established, the next step involves adding quantum errors to the noise model. Following the creation of a realistic quantum model with the noise components included, the pipeline proceeds to circuit parameter training with optimizers. Here, optimization algorithms are employed to adjust the parameters of the quantum circuits. The output of the training phase yields trained parameters, which are then used to measure accuracies and fit times. This final stage involves evaluating the performance of the quantum circuits. Accuracy metrics and fit times (the time taken to achieve the optimal parameter configuration) are measured to determine the efficacy and efficiency of the quantum model.

B. Optimizers

The optimizer's main objective is to adjust the parameters of the quantum circuits to reach minimum of the cost function. The cost function quantifies the difference between the quantum circuit's predicted outputs and the true sentiment labels in the training dataset.

The optimizer iteratively updates the circuit parameters. During each iteration, it calculates how much each parameter should change to minimize cost function. The size of each change is controlled by the learning rate in gradient-based optimizers or by step sizes in gradient-free optimizers. The way the optimizer updates the circuit parameters depends on the type of optimizer algorithm used. In this study, the optimizer algorithms are SPSA, Nelder-Mead, and Adam.

The optimization process continues until convergence criteria are met. These criteria might include a predetermined number of iterations, achieving a cost function value below a specific threshold, or observing negligible changes in the cost function value between iterations. In this study, the convergence criteria used is a predetermined number of iterations, set at 130.

V. Experimental Results

This section shows the experimental results of three optimization methods for identifying the sentiment of a sentence: SPSA, Nelder-Mead, and Adam optimizers. The steps are shown in Fig. 2. The aims are: 1) To see the effect of quantum errors on each algorithm. 2) To observe the effect of various combinations of quantum errors. 3) To find the suitable optimization algorithm for the task and data in a noisy environment.

In the experiment, we provides the total 290 sentences consisting 170 sentences for training, 50 sentences for development, and 60 sentences for test [17]. Each sentence consists of 4-5 words. We label the sentences and the label of 0 or 1 shows negative or positive sentiment. The sentences are based on 29 words in the restaurant domain as follows:

- The food/meal/service/staff was not [adjective].
- The food/meal/service/staff was [adjective].
- I like/love/dislike/hate the restaurant.
- The restaurant showed [adjective] service/staff.
- The restaurant had [adjective] food/meals/service/staff.
- I liked/loved/disliked/hated the [adjective] food/service.
- The restaurant cooked [adjective] food/meals.
- The restaurant was [adjective].
- I had [adjective] food/meals/service/staff.

In the experiment, we simulate the QNLP algorithm using AMD Ryzen 7 5000 processor and 16 GB of RAM on Windows 11 OS. We carried out two sets of experiments: experiments in a noiseless model and experiments in a noisy model. For experiments in the noisy model, we carried out three sets of quantum errors: depolarizing error, readout error, and both. Note that to ensure reproducibility, the same seed was used for each experiment. For simplicity, we only show the image for Adam optimizer, but the whole results are shown in Table I.

In this experiment, we use the best hyperparameters, and the training is conducted up to 130 iterations. Moreover, we compare the accuracy of all optimizers for noiseless and noisy experiments. As shown in Table I, by using Adam optimizer, the accuracy reaches up to 97% in noiseless environments and still reaches up to 70% in noisy environments.

A. Experiment Analysis

From the experiment's result, we can conclude that the Adam optimizer gives higher accuracy and outperforms SPSA and

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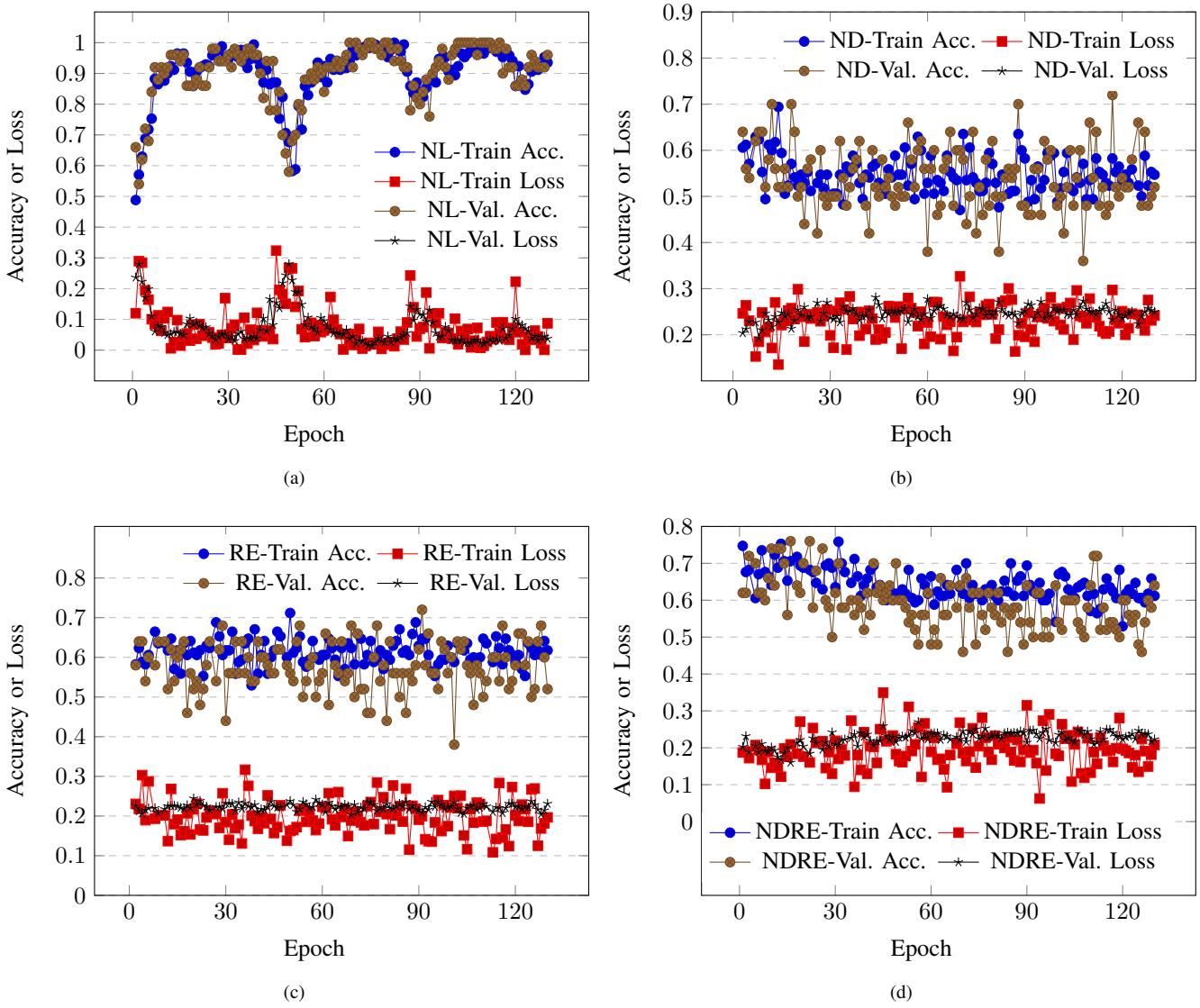


Fig. 3: Quantum Sentiment Training and Validation Simulations using Adam Optimizer in (a) Noiseless (NL), (b) Noisy Depolarization (ND), (c) Readout Error (RE), and (d) Both Noisy Depolarization and Readout Error (NDRE)

TABLE I: Experimental Results for Several Optimizers in Noiseless and Noisy Simulation

Optimizer	Noiseless (NL)	Noise Depolar. (ND)	Readout Error (RE)	Both (NDRE)
SPSA	83%	47%	52%	42%
Nelder-Mead	53%	48%	45%	52%
Adam	97%	65%	61%	70%

Nelder-Mead in both noiseless and noisy environments. Adam's adaptability helps it converge faster and more reliably than other methods, especially in high-dimensional spaces or when the objective function is noisy. In Fig. 3(a), a loss spike during noiseless training is observed, which occurs due to the nature of Adam's instability when the batch size is not sufficiently large [18]. From the noisy environment experiments in Figs. 3(b, c, d), it can be seen that the model does not increase much in accuracy over time and instead has a relatively stable accuracy for the training set and an unsteady accuracy for the development set.

The results from SPSA and Nelder-Mead models both have low accuracy in noisy environments, with SPSA having 42% accuracy and Nelder-Mead having 52% accuracy with readout and depolarizing error. Nelder-Mead tends to have low accuracy in high-dimensional spaces because it can struggle with the

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complexities of higher dimensions based on this paper [19].

Here is our analysis regarding why Adam optimizer is better than other optimizers, especially in noisy environments: 1) **Adaptive Learning Rates:** Adam's adaptive learning rate mechanism allows it to adjust to the varying noise levels caused by quantum errors. Depolarizing errors and readout errors can introduce fluctuations and inaccuracies in the gradient estimates. 2) **Gradient Handling:** Quantum errors can lead to noisy gradient estimates, where the gradients fluctuate around the true gradients due to error-induced variations. Adam optimizer, through its momentum and adaptive learning rate mechanisms, can smooth out these fluctuations over iterations. 3) **Robustness in High Dimensions:** Adam's robustness and efficiency in high-dimensional optimization scenarios allow it to navigate through the parameter space effectively, even when the objective function evaluations are noisy due to quantum errors. Methods like Nelder-Mead, which rely on simplex-based operations, may struggle more in such high-dimensional spaces affected by quantum noise. 4) **Convergence Speed:** Adam's efficient use of gradient information and adaptive nature generally leads to faster convergence compared to other methods like Nelder-Mead or SPSA, especially when dealing with noisy quantum environments.

VI. Conclusions and Future Works

The research demonstrates that the Adam optimizer outperforms SPSA and Nelder-Mead in QNLP simple sentiment analysis tasks, both in noiseless and noisy environments. In noiseless conditions, Adam achieved an accuracy of 97%, significantly higher than SPSA's 83% and Nelder-Mead's 53%, showcasing its superior performance in optimizing quantum circuits.

In noisy environments, Adam's robustness is further highlighted by its higher accuracy across various noise models. When both readout and depolarizing noise were present, Adam still led with 70%, while SPSA and Nelder-Mead were at 42% and 52%, respectively.

Future research could explore more adaptive strategies for handling noise in quantum sentiment analysis. Further investigation into other approaches for integrating Nelder-Mead, which shows subpar results in this study, with a more suitable quantum computing framework could also be pursued.

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